

Faculty of Science & Technology
PES's Modern College of Engineering
(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

Level 7: First Year M. Tech. (2024 Pattern)

Course Code	CSE08651
Course Name	Skill Development – II
Semester	IV
Experiment No.	6
Title	Interactive Research Dashboard with Streamlit

Problem Statement

To study and analyze the use of interactive visualization tools for presenting and exploring research data in an intuitive and user-friendly manner.

Objective

To make use of Streamlit for developing an interactive research dashboard for data visualization and analysis.

Outcome

CO653.6: Design and develop interactive dashboards for effective research data visualization and analysis.

Software / Hardware Requirements

- Streamlit Framework
- Python Programming Environment
- Data Visualization Libraries (Matplotlib / Plotly)
- Web Browser
- Personal Computer or Laptop

Theory

Interactive data visualization plays a crucial role in modern research by enabling researchers to explore datasets dynamically and gain deeper insights. Traditional static visualizations often limit user interaction and restrict exploratory analysis. Interactive dashboards address these limitations by allowing users to filter, zoom, and manipulate data in real time.

Streamlit is an open-source Python framework designed for building interactive web applications for data science and research. It allows researchers to convert Python scripts into shareable web applications with minimal effort. Streamlit supports various data visualization libraries and provides interactive components such as sliders, dropdowns, and buttons, which enhance user interaction.

By using Streamlit, researchers can develop dashboards that present research data, models, and results in an intuitive manner. These dashboards facilitate better understanding of experimental outcomes, support exploratory data analysis, and improve the communication of research findings to both technical and non-technical audiences.

Steps Involved

1. Streamlit was installed in the Python environment.
2. A Python script was created for the research dashboard.
3. Research data was loaded and processed within the application.
4. Interactive visualizations were implemented using Streamlit components.
5. The dashboard was executed and tested using the Streamlit framework.

Results

An interactive research dashboard was successfully developed using Streamlit. The dashboard enabled dynamic data visualization and improved understanding of research data.

Analysis and Conclusion

Analysis: Streamlit enabled efficient development of an interactive dashboard, improving data exploration and user engagement compared to static visualizations.

Conclusion: Interactive dashboards developed using Streamlit enhance research data analysis and presentation. Their use supports effective visualization and communication of research outcomes.

References

1. Streamlit, “Streamlit Documentation,” Available: <https://docs.streamlit.io>
2. McKinney, W., *Python for Data Analysis*, O’Reilly Media.